Lecture 14: P-N junction, diodes, transistors

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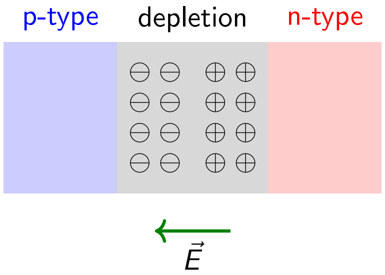
Remember: P-N junction is all about forming a depletion region

- Region near junction **depleted** of mobile carriers
- Fixed positive charges (N_D^+) on n-side, negative charges (N_A^-) on p-side
- Space charge creates internal electric field pointing from n to p
- Depletion region is charged because p-type and n-type semiconductors are **neutral**
- Why isn't the depletion region the entire semiconductor? $\rightarrow$ **electric field**
- How to find the electric field? $\rightarrow$ **Gauss' law**



Last lecture: Poisson equation in depletion region

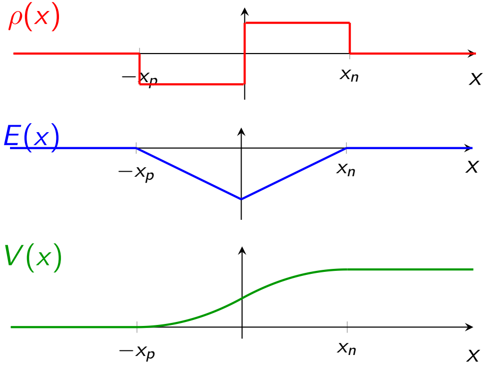
- Charge density in depletion approximation:

$$\rho(x) = \begin{cases} -eN_A & -x_p < x < 0 \\ +eN_D & 0 < x < x_n \end{cases}$$

- Poisson equation:

$$\frac{d^2V}{dx^2} = -\frac{\rho}{\varepsilon}$$

- Integrate once $\Rightarrow E(x)$ linear
- Integrate twice $\Rightarrow V(x)$ quadratic
- Boundary conditions: $E = 0$ and V continuous at $\pm x_{p,n}$



Last lecture: built-in voltage

Concentration of carriers as a function of temperature and dopant concentration:

$$\frac{n_n}{n_p} = \frac{N_A N_D}{n_i^2} = \exp\left(\frac{eV_{bi}}{k_B T}\right)$$

Solve for the built-in voltage V_{bi}:

$$\boxed{V_{bi} = \frac{k_B T}{e} \ln\left(\frac{N_A N_D}{n_i^2}\right)}$$

This thermodynamic expression must equal the electrostatic result:

$$V_{bi} = \frac{e}{2\varepsilon}\left(N_A\, x_p^2 + N_D\, x_n^2\right)$$

Equating the two gives a relation between the depletion widths x_p, x_n and the doping concentrations N_A, N_D.

Conceptual check: P-N junction basics

1. A Si p-n junction has $N_A = 10^{17}\,\mathrm{cm}^{-3}$ and $N_D = 10^{16}\,\mathrm{cm}^{-3}$. On which side is the depletion region wider? By what factor?
2. At equilibrium, is there a current flowing through the junction? Is there a diffusion flux? Is there a drift flux? Explain.
3. Compute V_{bi} for the junction above at $T = 300\,\mathrm{K}$. Use $n_i = 1.5 \times 10^{10}\,\mathrm{cm}^{-3}$.
4. If the temperature increases, does V_{bi} increase or decrease? Consider what happens to $n_i(T)$.

Conceptual problems: non-trivial to try at home

1. Starting from charge neutrality ($N_A x_p = N_D x_n$) and the electrostatic expression for V_{bi}, show that the total depletion width is:

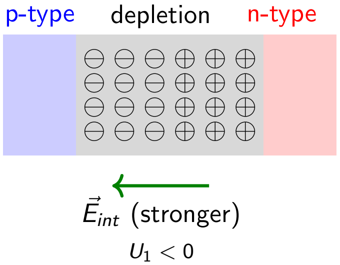
$$W = x_p + x_n = \sqrt{\frac{2\varepsilon V_{bi}}{e}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)}$$

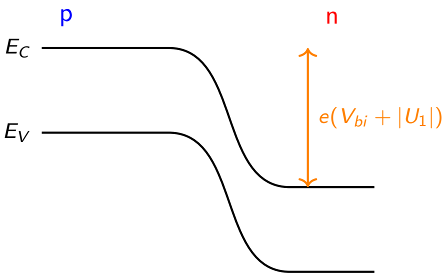
2. A p-n junction has $N_A = N_D = 10^{16}\,\mathrm{cm}^{-3}$. A second junction has $N_A = 10^{18}$ and $N_D = 10^{14}\,\mathrm{cm}^{-3}$. Compare V_{bi} for both. Which has a wider depletion region? In the asymmetric case, which side hosts nearly all of it?

3. The peak electric field is $E_{\max} = eN_A x_p/\varepsilon$. For the asymmetric junction above, estimate $E_{\max}$ and W. Use $\varepsilon = 12\,\varepsilon_0$.

Reverse bias: barrier increases, depletion widens

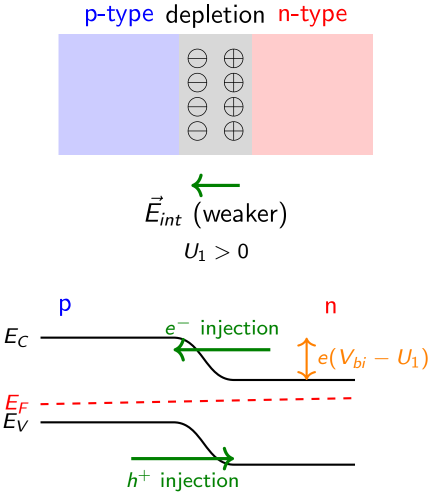
- Applied voltage $U_1 < 0$ adds to built-in potential
- Barrier becomes $e(V_{bi} + |U_1|)$
- Depletion region **widens**: fewer carriers cross the barrier
- The internal electric field $\vec{E}_{int}$ becomes stronger





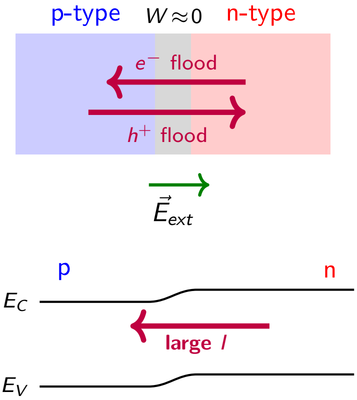
Forward bias: barrier decreases, depletion narrows

- Applied voltage $U_1 > 0$ opposes built-in potential
- Barrier reduced to $e(V_{bi} - U_1)$
- Depletion region **narrows**: majority carriers injected
- The internal electric field $\vec{E}_{int}$ becomes weaker
- Large diffusion current increases exponentially with U_1



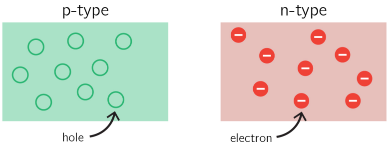
Strong forward bias: ohmic conduction regime

- Applied voltage $U_1 \to V_{bi}$: barrier nearly eliminated
- Depletion region **collapses**: $W \to 0$
- Massive injection of majority carriers across junction
- The internal electric field $\vec{E}_{int}$ is negligible
- The external electric field $\vec{E}_{ext}$ drives conduction of electricity

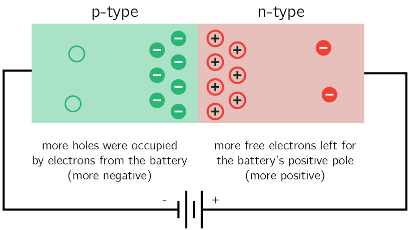


Summary of P-N junctions

prior to contact

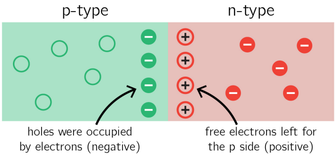


reverse bias

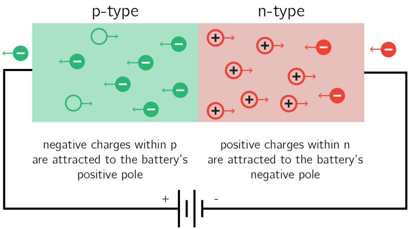


reverse bias = no net current

after contact



forward bias



forward bias = net current

Derivation: Shockley diode equation (1/4)

Why exponential? Carriers follow Boltzmann statistics. The fraction of carriers with energy $> \Delta E$ is $\propto e^{-\Delta E/k_B T}$.

At equilibrium ($U_1 = 0$), the barrier is eV_{bi}. Diffusion current:

$$I_{\text{diff}}^{(0)} \propto \exp\left(-\frac{eV_{bi}}{k_B T}\right)$$

This is exactly balanced by drift of minority carriers across the junction:

$$I_{\text{drift}} = I_{\text{diff}}^{(0)} \equiv I_0 \qquad \Longrightarrow \qquad I_{\text{net}} = 0$$

Derivation: Shockley diode equation (2/4)

Under bias, the barrier changes to $e(V_{bi} - U_1)$:

$$I_{\text{diff}} \propto \exp\left(-\frac{e(V_{bi} - U_1)}{k_B T}\right) = \underbrace{\exp\left(-\frac{eV_{bi}}{k_B T}\right)}_{\propto I_0} \cdot \exp\left(\frac{eU_1}{k_B T}\right)$$

Therefore:

$$\boxed{I_{\text{diff}} = I_0 \exp\left(\frac{eU_1}{k_B T}\right)}$$

Physical intuition

Lowering the barrier by eU_1 exponentially increases the number of carriers with enough thermal energy to cross.

Derivation: Shockley diode equation (3/4)

Drift current is bias-independent (for moderate voltages):

- Minority carriers that wander into the depletion region are swept across by $\vec{E}$
- This depends on the *supply* of minority carriers, not the barrier height
- Therefore $I_{\text{drift}} \approx I_0$ regardless of U_1

Net current = what goes over the barrier − what is swept back:

$$I = I_{\text{diff}} - I_{\text{drift}} = I_0 \exp\left(\frac{eU_1}{k_B T}\right) - I_0$$

Derivation: Shockley diode equation (4/4)

Rectifier (Shockley) Equation

$$\boxed{I = I_0 \left[\exp\left(\frac{eU_1}{k_B T} \right) - 1 \right]}$$

Limiting cases:

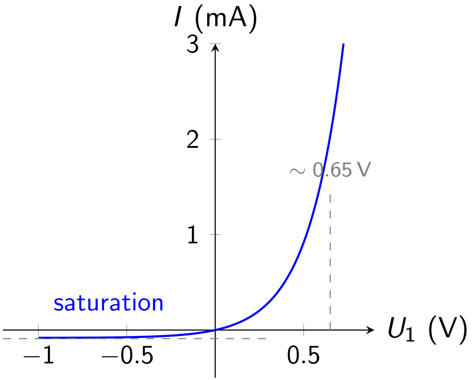
$$U_1 \gg \frac{k_B T}{e}: \quad I \approx I_0 \, e^{eU_1/k_B T} \quad \text{(exponential growth)}$$

$$U_1 \ll -\frac{k_B T}{e}: \quad I \approx -I_0 \quad \text{(reverse saturation)}$$

Exponential I-V characteristic of the diode

- At 300 K: $k_B T/e \approx 26\,\mathrm{mV}$
- Current increases exponentially
- "Turn-on" voltage for Si $\approx$ 0.6–0.7 V
- Reverse bias: current saturates at $-I_0$ (typically nA–μA for Si)
- $I_0 \propto n_i^2 \propto \exp(-E_g/k_B T)$: strong temperature dependence

$$I = I_0 \left[\exp\left(\frac{eU_1}{k_B T}\right) - 1\right]$$



Conceptual problems: Bias and the Shockley equation

1. At equilibrium, $I_{\text{diff}} = I_{\text{drift}} = I_0$. Under reverse bias, does I_{diff} increase or decrease? What about I_{drift}? Explain physically why the net current saturates at $-I_0$.

2. A Si diode at 300 K has $I_0 = 10\,\text{nA}$. Compute the current at $U_1 = 0.5\,\text{V}$ and at $U_1 = 0.6\,\text{V}$. Verify by how much did the current increase.

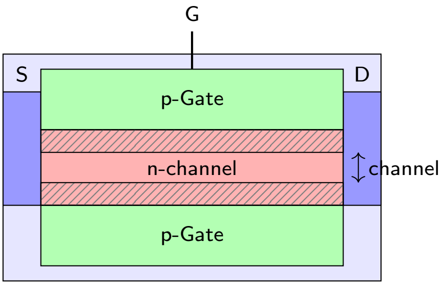
3. The temperature is raised from 300 K to 350 K. Qualitatively, what happens to I_0, V_{bi}, and the turn-on voltage? Which effect dominates: the T prefactor in $k_B T/e$ or the exponential increase of $n_i^2(T)$?

4. In the strong forward bias regime ($U_1 \to V_{bi}$), the Shockley equation predicts $I \to \infty$. What limits the current in a real device? Sketch how the I-V curve deviates from the ideal exponential at high forward bias.

Making a device with two P-N junctions: the junction field-effect transistor (J-FET)

Operating principle

- Gate voltage modulates depletion width
- Reverse gate bias widens depletion
- Conducting channel narrows
- High input impedance (no gate current)



More negative V_{GS} $\rightarrow$ wider depletion $\rightarrow$ narrower channel $\rightarrow$ less I_D

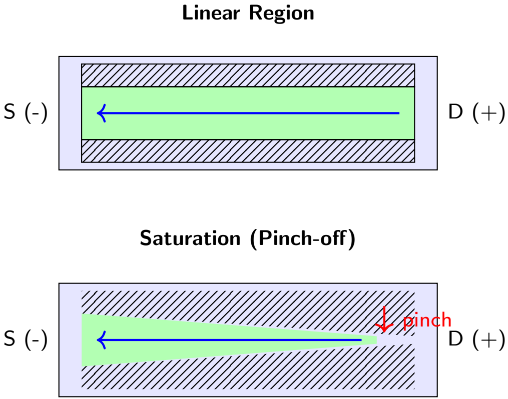
Core idea of a transistor

We can use a voltage to control the conductivity of a material!

J-FET saturation: channel pinch-off mechanism

Pinch-off

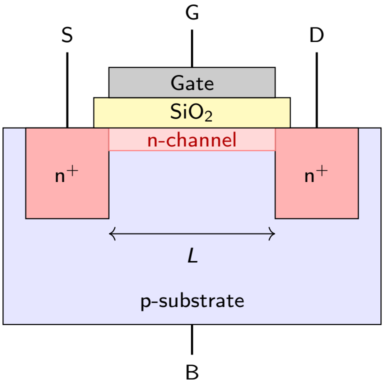
- Increasing the depletion zone reduces the conductivity of the channel
- At saturation, the depletion region completely *pinches off* the channel.
- **Pinch-off:** $I_{DS} \to 0$
- The geometry of the pinch-off is due to the polarity of the V_{DS}



The Metal-Oxide-Semiconductor FET (MOSFET)

Operating principle

- Gate separated from channel by thin oxide (SiO_2)
- $V_{GS} > V_{thresh}$: electric field attracts carriers, creating conducting channel



Larger $V_{GS} \rightarrow$ more carriers in channel $\rightarrow$ more I_{DS}

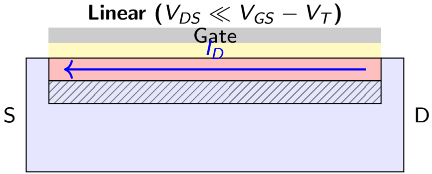
Core idea

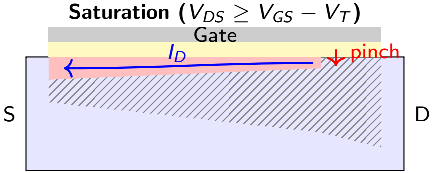
The gate *creates* the channel via electric field — no current needed!

MOSFET saturation: channel pinch-off mechanism

Pinch-off

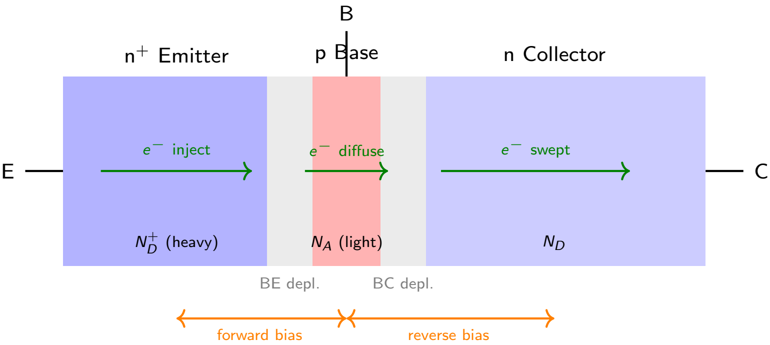
- In the linear region, the carrier layer is roughly uniform along the channel
- Increasing V_{DS} reduces the gate-to-channel voltage near the drain
- At $V_{DS} = V_{GS} - V_{thresh}$: inversion layer vanishes at drain end $\rightarrow$ **pinch-off**





Bipolar junction transistor: controlling currents with currents

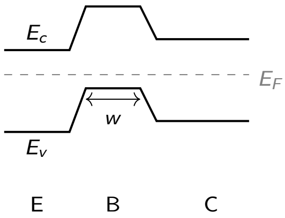
- **Emitter** (heavily doped): injects minority carriers into base region
- **Base** (thin, lightly doped): most injected carriers diffuse through without recombining
- **Collector** (reverse-biased): sweeps minority carriers arriving from base



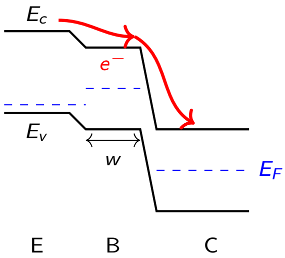
BJT: controlling current through the application of a bias

- Forward-biased E-B junction injects electrons into p-type base
- Thin base ($W \ll L_n$) ensures most electrons reach collector before recombining
- Small base current (recombination) controls large current I_{EC}

No bias (equilibrium)



Active mode bias



Metal-semiconductor junctions

- Metal work function W: energy from E_F to vacuum; semiconductor affinity χ: energy from E_c to vacuum
- Contact $\rightarrow$ electrons transfer from SC to metal until E_F aligns
- Charge transfer creates depletion layer $\rightarrow$ electric field $\rightarrow$ **band bending**
- Schottky barrier height: $V_B = W - \chi$;